



Faculty of Technology and Engineering

Computer Science and Engineering

Practical

Academic Year	:	2025-26	Semester	:	6
Course code	:	CSE312	Course name	:	Design of language processing

Practical - 9

1. Objective:

Develop a YACC program to validate input strings based on the given grammar. The program should parse the string using the grammar rules and determine whether the string is valid or invalid.

$$S \rightarrow i E t S S' \mid a$$

$$S' \rightarrow e S \mid \varepsilon$$

$$E \rightarrow b$$

2. Program Code:

```
#include <iostream>
#include <string>
using namespace std;

string input;
int pos;

// Forward declarations
bool parseS();
bool parseSPRime();
bool parseE();

// Peek at current character (skip spaces)
char peek() {
    while (pos < (int)input.size() && input[pos] == ' ')
        pos++;
    if (pos >= (int)input.size()) return '\0';
    return input[pos];
}

// Consume and return current character
char consume() {
    while (pos < (int)input.size() && input[pos] == ' ')
        pos++;
}
```

```

    if (pos >= (int)input.size()) return '\0';
    return input[pos++];
}

// E -> b
bool parseE() {
    if (peek() == 'b') {
        consume();
        return true;
    }
    return false;
}

// S' -> e S | ε
bool parseSPrime() {
    if (peek() == 'e') {
        consume(); // consume 'e'
        return parseS();
    }
    // ε production — always succeeds (S' can be empty)
    return true;
}

// S -> i E t S S' | a
bool parseS() {
    char c = peek();

    if (c == 'i') {
        consume(); // consume 'i'
        if (!parseE()) return false;
        if (peek() != 't') return false;
        consume(); // consume 't'
        if (!parseS()) return false;
        return parseSPrime();
    } else if (c == 'a') {
        consume(); // consume 'a'
        return true;
    }
    return false;
}

int main() {
    string line;
    if (!getline(cin, line)) {
        cerr << "No input provided." << endl;
        return 1;
    }

    input = line;
    pos = 0;

```

```

bool valid = parseS();

// After parsing S, we must have consumed all input
// (skip trailing spaces)
while (pos < (int)input.size() && input[pos] == ' ')
    pos++;

if (valid && pos == (int)input.size()) {
    cout << "Valid string" << endl;
} else {
    cout << "Invalid string" << endl;
}

return 0;
}

```

3.Output:

```

Last login: Sat Mar 28 19:23:06 on ttys000
debdootmanna@Debdoots-MacBook-Air DLP % g++ -o 9 9.cpp
debdootmanna@Debdoots-MacBook-Air DLP % echo "ibtaea" | ./9
Valid string
debdootmanna@Debdoots-MacBook-Air DLP % echo "ibtai" | ./9
Invalid string
debdootmanna@Debdoots-MacBook-Air DLP % echo "ibtibta" | ./9
Valid string
debdootmanna@Debdoots-MacBook-Air DLP % echo "ibtb" | ./9
Invalid string
debdootmanna@Debdoots-MacBook-Air DLP % 

```